1.3 Sequence and Limit

A countable collection $\{a_n\}_{n=1}^{\infty}$ where $a_n \in R$ is called a sequence of real numbers.

We say $\{a_n\}$ has a limit a, denoted by $\lim_{n\to\infty} a_n = a$, or $a_n \to a$, if for any $\epsilon > 0$, there is a N such that for any $n > N$ we have $|a_n - a| < \epsilon$.

Proposition 1.3.1: If $\{a_n\}$ has a limit, the limit is unique. *Proof:* Suppose b_1 are b_2 are limits of $\{a_n\}$. We show that $b_1 = b_2$. Suppose $b_1 \neq b_2$. Without loss of generality, we assume $b_1 < b_2$. Let $\epsilon = \frac{b_2 - b_1}{4}$. Then there is a N_1 such that when $n > N_1$, $a_n < b_1 + \epsilon$. There is a N_2 such that when $n > N_2$, $a_n > b_2 - \epsilon > b_1 + \epsilon$. So take $N = \max\{N_1, N_2\}$. When $n > N$, we must have both $a_n < b_1 + \epsilon$ and $a_n > b_1 + \epsilon$, which is a contradiction. **QED**

The followings theorems involve sequence and limit and are also three fundamental theorems of real numbers.

We say a sequence $\{a_n\}$ is **bounded** if there is $c \geq 0$ such that $|a_n| \leq c$ for all n.

We say a sequence $\{a_n\}$ is **increasing** if $a_n \leq a_{n+1}$ for all n. If $a_n < a_{n+1}$ for all n, we say it is **strictly increasing**.

We say a sequence $\{a_n\}$ is **decreasing** if $a_n \geq a_{n+1}$ for all n. If $a_n > a_{n+1}$ for all n, we say it is **strictly decreasing**.

We say a sequence $\{a_n\}$ is **monotone** if it is either increasing or decreasing. We say it is **stricly monotone** if it is either strictly increasing or strictly decreasing.

Theorem 1.3.2 (Monotone Convergence Theorem) : A monotone sequence converges, if and only if, it is bounded. *Proof:* Let $\{a_n\}$ be a sequence. Suppose $\{a_n\}$ converges to a. Consider a $\epsilon > 0$. Then there is N such that $|a_n - \epsilon| < a$ for $n > N$. Thus, $|a_n| < |a| + |a_n - a| < |a| + \epsilon$. Take $c = \max\{|a_1|, \ldots, |a_{N-1}|, |a| + \epsilon\}$, and clearly $|a_n| \leq c$ for all n. Hence, $\{a_n\}$ is bounded.

Conversely, suppose $\{a_n\}$ is monotone and bounded. We may assume $\{a_n\}$ is increasing and upper bounded. By Completeness Axiom, $\{a_n\}$ has a supremum a. We show that a is the desired limit.

To this end, note that since a is the supremum, for any $\epsilon > 0$, $a - \epsilon$ is not a upper bound of a_n. Then there exists a_N such that $|a_N - a| < \epsilon$. Since $\{a_n\}$ is increasing, for all $n > N$ we must have $|a_n - a| < \epsilon$. Therefore, $a_n \to a$. **QED**

Theorem 1.3.3 (Bolzano-Weierstrass Theorem): Every bounded sequence has a convergent subsequence. *Proof:* Let $\{a_n\}$ be a bounded sequence. We denote by E_k the set $\{a_n\}_{n\geq k}$. Consider the closure $F_k = \overline{E_k}$. Since $\{a_n\}$ is bounded, each F_k is also bounded. Then $F_1, \ldots, F_k, \ldots$ is a descending sequence of non-empty closed and bounded sets. By Nested Set Theorem, there exists $a \in \bigcap F_k$. We construct a subsequence $\{a_{n_j}\}$ that converges to a as follows.

Note a is a point of closure of any E_k. So take $c_1 = 1$. There exists $a_{n_1} \in E_1$ such that $|a_{n_1} - a| < c_1$. Take $c_2 = \frac{1}{2}$. There exists $a_{n_2} \in E'$, where E' is some E_k that does not contain a_{n_1} and $n_2 > n_1$, such that $|a_{n_2} - a| < c_2$. Continuing this process, we take $c_j = \frac{1}{j}$. There exists a_{n_j} such that $n_j > n_i, i = 1, \ldots, j-1$ and $|a_{n_j} - a| < \frac{1}{j}$. The resulting subsequence $a_{n_1}, \ldots, a_{n_j}, \ldots$ clearly converges to a. **QED**

We say a sequence $\{a_n\}$ is a Cauchy sequence, if for any $\epsilon > 0$ there exists a N such that when $m, n > N$, $|a_m - a_n| < \epsilon$.

Theorem 1.3.4 (Cauchy Convergence Theorem): A sequence coverges if and only if it is a Cauchy sequence. *Proof:* Let $\{a_n\}$ be a sequence. If $a_n \to a$, then for any $\epsilon > 0$ there is a N such that $|a_n - a| < \frac{\epsilon}{2}$ when $n > N$. So for all $m, n > N$, we have
$|a_m - a_n| = |(a_m - a) - (a_n - a)| < |a_m - a| + |a_n - a| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$. Thefore, $\{a_n\}$ is a Cauchy sequence.

Conversely, suppose $\{a_n\}$ is a Cauchy sequence. It is not difficult to show that $\{a_n\}$ is bounded. By Bolzano-Weierstrass Theorem, there is a convergent subsequence $a_{n_k} \to a$. We claim $a_n \to a$. To this end, consider any $\epsilon > 0$. There exists a N_1 such that $|a_{n_k} - a| < \frac{\epsilon}{2}$ for all $n_k > N_1$. There exists a N_2 such that $|a_n - a_{n_k}| < \frac{\epsilon}{2}$ for all $n, n_k > N_2$. Now take $N = \max\{N_1, N_2\}$. When $n > N$, we have

$$|a_n - a| < |a_n - a_{n_k}| + |a_{n_k} - a| < \epsilon$$

. Therefore, $a_n \to a$. **QED**

Linearity and Monotonity

Proposition 1.3.5: $\{a_n\}$ and $\{b_n\}$ are two convergent sequences.

- (a) For any α, β, $\lim_{n\to\infty}(\alpha \cdot a_n + \beta \cdot b_n) = \alpha \cdot \lim_{n\to\infty} + \beta \cdot \lim_{n\to\infty} b_n$
- (b) If $a_n \leq b_n$ for all n, then $\lim_{n\to\infty} a_n \leq \lim_{n\to\infty} b_n$

Proof: (a) Let $a = \lim_{n\to\infty} a_n$ and $b = \lim_{n\to\infty} b_n$. We may assume $\alpha \neq 0$ and $\beta \neq 0$.

For any $\epsilon > 0$, there exists N_1 such that when $n > N_1$, $|a_n - a| < \frac{\epsilon}{|\alpha|}$. There exists N_2 such that when $n > N_2$, $|b_n - b| < \frac{\epsilon}{|\beta|}$. Take $N = \max\{N_1, N_2\}$. When $n > N$, we have
$|(\alpha \cdot a_n + \beta \cdot b_n) - (\alpha \cdot a + \beta \cdot b)| \leq |\alpha| \cdot |a_n - a| + |\beta| \cdot |b_n - b| < \epsilon$.

(b) Supppose $a > b$. Take $\epsilon = \frac{a-b}{4}$. There exists a N such that when $n > N$,

$$b_n < b + \epsilon < a - \epsilon < a_n$$

, which is a contradiction. **QED**

We say a sequence $\{a_n\}$ converges to $+\infty$ if for any $c > 0$ there exists a N such that when $n > N$, $a_n > c$. We denote such a convergence by $\lim_{n\to\infty} a_n = \infty$. The meaning of

$$\lim_{n\to\infty} a_n = -\infty$$

is self-explanatory.

We define

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup\{a_k | k \geq n\}$$

. Similarly, we define

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf\{a_k | k \geq n\}$$

.

The $\limsup$ and $\liminf$ involve some properties that appera infnitely often in the sequence.

Note that $b_n = \sup\{a_k | k \geq n\}$ is a **decreasing** sequence. And $c_n = \inf\{a_k | k \geq n\}$ is an **increasig** sequence.

Proposition 1.3.6: $\{a_n\}$ and $\{b_n\}$ are two sequences.

- (a) $\limsup a_n = a \in R$ if and only if for any ϵ there are infinitely many $a_n > a - \epsilon$ and finitely many $a_n < a + \epsilon$.
- (b) $\limsup a_n = \infty$ if and only if for any $c > 0$ there are infinitely many $a_n > c$.
- (c) $\limsup a_n = -\liminf(-a_n)$
- (d) $\lim a_n = a$ if and only if $\limsup a_n = \liminf a_n = a$
- (e) If $a_n \leq b_n$ for all n, then $\limsup a_n \leq \liminf b_n$

Proof: The proof is simple and omitted. **QED**